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ROTOR X

REDSEA STEM SCHOOL

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CHAPTER I

Present and Justify a Problem and Solution Requirements: -

EGYPT'S GRAND CHALLENGES

1. Improve the use of alternative energies.

2. Recycle garbage and waste for economic and environmental purposes

3. Address & reduce pollution in air, water, and soil.

4. Reduce and adapt to the effects of climate change.

Improve the use of alternative energies:

Overall:

Egypt is a place for renewable energy because it has a lot of land and a sunny climate. The government of Egypt wants to generate much of its electricity from alternative sources such as wind and solar energy. They want to get 42% of their electricity from alternative sources by the year 2030 and more than 60% by the year 2040.


Figure 1

They will use wind, solar, and hydropower to do this. The European Union is helping Egypt plan for its energy. Companies are also playing a role in this. Egypt is making deals with these companies to buy the energy they produce.


Figure 2


Figure 3

Egypt is also working to connect its energy system with neighboring countries' systems. Egypt is trying to make its energy grid smarter so it can handle all the new energy. The country also aims to boost local content in large projects and has a priority on electric vehicles, though investment is limited by currency issues.

Causes*:*

1. Available everywhere:

A lot of countries must import fossil fuels from other places. The thing is that every country has its own renewable sources, like solar and wind power. These renewable sources of energy can be used to make most of the electricity we need by the year 2050. If we use sources, we will not have to rely on other countries for fossil fuels. This will make our countries stronger. Our economy will do better. Renewable energy is a thing for every country because it is available everywhere. We can use energy to make electricity and reduce our dependence on fossil fuels.

2. Cheaper than fossil fuels:

The cost of technology is going down really fast. Now, 90 percent of new renewable technology projects are cheaper than fossil fuel alternatives. Renewable technologies have the potential to supply most of the world's electricity in the coming decades. This is because the cost of ***technology*** is falling quickly, and renewable technologies are getting better all the time.

3. Healthier for people:

Air pollution from fuels is really bad for people. It causes a lot of deaths every year. On the other hand, clean energy sources, like wind and solar, are very good. Wind and solar reduce pollution. They make public health better. This is because wind and solar are energy sources that do not produce air pollution like fossil fuels do. Fossil fuels are a problem, and wind and solar can help solve this problem.

4. Creates more jobs:

The clean energy sector gives jobs to a lot of people, millions actually. It creates three times more jobs for every dollar that is put into it compared to the fossil fuel industry. We can expect the clean energy sector to have a lot of jobs in the future. The clean energy sector is ***in*** 2022***;*** the country made many achievements and reached an impressive total renewable energy capacity of 6,322 MW. Hydropower stands as one of the cornerstones of Egypt's energy strategy, with a robust capacity of 2,832 MW. The historic Aswan High Dam epitomizes Egypt's experience in the harnessing of water resources towards the generation of power and thus contributes to the national grid significantly, really going to grow. This means a lot of new jobs will be created in the clean energy sector.

Impact:

Solar energy in Egypt has also experienced a remarkable upsurge, with capacity reaching 1,724 MW. Leveraging its geographical advantage of more than 2,000 kWh/ (m^2 year) of solar irradiance, Egypt is an ideal region for both photovoltaic (PV) and concentrated solar power (CSP) projects.

Another area where Egypt is making remarkable progress is in wind energy. Egypt has reached a capacity of 1,643 MW in this sector. The high average speed of winds in the Gulf of Suez provides it with suitable conditions to establish wind farms to generate electricity.

The Egyptian government has ambitious plans to increase the percentage of electricity generated from renewable resources in the country by the year 2035 to reach 42%. For instance, in the fiscal year 2023-2024, the Egyptian government is planning to invest an amount of US $2.6 billion in the renewable energy sector to increase its contributions to the Egyptian energy sector by 11.8%. The Egyptian government’s focus on renewable energy can be seen as a strategic step towards the sustainability and development of the Egyptian energy sector. The developments in the Egyptian renewable energy sector are a beacon of hope for environmental sustainability globally.

Waste Management and Recycling in Egypt

1. Introduction

Managing waste in Egypt has become a pressing engineering problem due to the fast population growth and rapid urbanization. The millions of tons of waste produced every year exert tremendous strain on infrastructure, the environment, and the health of citizens. The ineffective management of waste causes significant harm to the country and may lead to various negative consequences in the future.

To solve this problem, the author suggests considering an engineering approach to waste management that involves recycling of materials to build a hybrid wind turbine. Thus, the author aims to combine waste management and renewable energy technologies in one project.

2. Hazardous Waste Management

The presence of harmful chemicals, heavy metals, and industrial waste products poses great risks if not treated adequately. This hazardous material can be released into the air, groundwater, and soil, causing significant environmental issues.

While the suggested project cannot be used for managing hazardous waste, it promotes:

- Decreasing the usage of newly produced items;

- Increasing the consumption of environmentally friendly materials;

- Developing sustainable engineering practices.

Such steps are needed to reduce further environmental problems at their sources.

3. Solid Waste Management:

The solid waste in the form of plastics, packaging, and scrap products creates many problems related to water and air pollution in Egypt. Moreover, this type of waste often blocks drainage networks, creating additional problems.

The current project can help manage solid waste as follows:

- Using plastic and wooden waste as building materials;

- Making mechanical parts out of waste materials;

- Avoiding unnecessary accumulation of waste in landfills.

This task becomes part of the engineering project objectives.

4. Recycling as a Sustainable Engineering Strategy:

In addition to solving waste management problems, recycling proves to be a great design practice in the current project. As demonstrated by the created prototype, waste materials can be transformed into efficient engineering elements.

Among the key applications of this design are the following:

- Using recycled plastic for turbine blades

- Applying recyclable materials for the turbine's base and its support system

- Minimizing the material cost and preserving the performance

In addition to reducing the amount of energy spent on material extraction, this approach will ensure sustainable engineering practices.

5. Environmental Impact

The use of recycled materials in the project will have a positive effect on the environment by:

- Minimizing solid waste accumulation

- Decreasing pollution caused by improper waste management

- Reducing greenhouse gas emissions during manufacturing

Moreover, when combined with renewable energy sources, recycling produces two benefits at once:

- Minimization of the solid waste problem

- Generation of clean energy

6. Economic Impact

Recycled materials' use in engineering systems implies such advantages as:

- Cost-effective production

- Reduction of reliance on expensive imported materials

- Low-cost scalability of the technology (concepts: scalable; at a low cost)

The example of the created prototype proves that sustainable energy systems can be developed despite their limited cost, thus increasing their scalability potential.

7. Social Impact

Recycling and reuse of materials have some positive impacts on society:

- Innovation and use of engineering solutions based on locally available materials

- Increase in the level of ecological awareness

- Development of sustainability practices within educational programs and industries

Furthermore, this project can serve as a great example of an application of engineering skills to solve a real-world problem with the use of accessible materials for the students.

8. Existing Challenges

Although the benefits of recycling seem evident, there are still many obstacles related to this process:

- Lack of infrastructure needed for collecting and sorting materials

- Limited awareness of the engineering use of recycled materials

- Possible health hazards due to improper recycling methods

The project helps tackle these challenges through an efficient and healthy implementation of recycled materials within an energy system.

9. Future Directions: Circular Economy

Transition to the circular economy concept will help to achieve sustainable development in the future. In the circular economy, waste is considered a valuable resource instead of being thrown away.

The presented project helps in the realization of this concept through the following ways:

- Reusing materials in engineering systems

- Minimizing waste production

- Increasing the efficiency of material use

By integrating recycling with renewable energy, the project contributes to a sustainable system that addresses environmental, economic, and social challenges simultaneously.

Address and reduce pollution fouling our air, water, and soil:

Overall:

Egypt is facing severe climate and environmental challenges that threaten its air, water, land, and ecosystems. Urbanization, population growth, industrialization, and overdependence on fossil fuels have sped up pollution and the rate of climate change effects.

More heat, unpredictable rainfall, and rising sea levels exacerbate air pollution, water scarcity, and land degradation problems. Egypt's reliance on the Nile River, with high coastal population density in the Nile Delta, makes it most vulnerable to these impacts.

For mitigation of such crises, Egypt has initiated various mitigation and adaptation strategies, including investment in renewable energy, public transport, integrated water management, and climate-resilient agriculture. However, these have to be scaled up and strengthened by professionals to prevent extreme economic, health, and social effects over the coming decades.

Cause:

1. Air Pollution:

a. Predominance of road transport, especially old vehicles and taxis (cause of over 80% congestion emissions).

b. Agricultural burning of rice straw produces an annual "black cloud" over the Nile Delta and Cairo.

(Figure 4)

Smog around Shanghai Tower

c. Industrial pollution, cheap subsidized fuel, and inefficient traffic management further contaminate the air.

2. Water Pollution:

a. Egypt's main water source, the Nile River, is polluted by industrial waste, fertilizers, sewage, and oil.

b. Dumping of waste in the river and washing animals in it contaminates supplies and spreads disease.

c. Climate change alters precipitation and reduces river flow, increasing competition for limited water.

(Figure 5)

Sewage thrown into a river

3. Soil Contamination and Degradation:

a. Abuse of chemical fertilizers and pesticides, and irrigation abuse, depletes soil fertility.

b. Land reclamation, urbanization, and salinization caused by seawater intrusion reduce farm productivity.

c. These render food security and economic growth susceptible.

(Figure 6)

A landfill

4. Climate Change Factors:

a. Overdependence on fossil fuels as energy and transport contributes to national emissions of CO_2, CH_4, and N_2O.

b. Ineffective irrigation and water-intensive crop cultivation worsen water scarcity.

c. Wetland and natural coastal buffers loss through development enhances storm and sea-level rise vulnerability.

Impact:

1. Environmental and Health Impact

a. Air pollution makes millions of Egyptians visit medical care each year for respiratory and cardiovascular disease.

b. Increased heat waves, droughts, and water pollution enhance disease epidemics and stress on health.

c. Loss of biodiversity and deteriorating air and water quality endanger ecosystems and human health.

2. Agricultural and Economic Impact:

a. Crop productivity (particularly wheat and maize) is expected to fall as a result of increased temperatures, salinity, and water scarcity.

b. Soil degradation weakens Egypt's food security and farmer incomes.

Egypt may lose GDP and lower productivity in agriculture, tourism, and industry without effective adaptation.

3. Coastal and Urban Impact:

a. Sea level rise and saltwater intrusion threaten millions along the Nile Delta, leveling homes, farmland, and infrastructure.

b. Urban sprawl and pollution render huge cities like Cairo less livable and more vulnerable to climate pressure.

c. Poor populations and slums are most vulnerable due to few resources and poor infrastructure.

4. Social and Institutional Impact:

a. Greater inequality as poor populations, especially rural peasants and urban poor, bear the greatest brunt.

b. Strain on healthcare, water management, and disaster response infrastructure.

c. Requirement of imminent policy compliance, education, and green technology to create long-term resilience.

Reduce and adapt to the effects of climate change

Overall:

Climate change: 3.6 billion people, nearly half the world's population, are currently exposed to its effects, including droughts, floods, and storms. This number will continue to rise as global temperatures do.

These developments are among the most serious risks we face today. They are wide-ranging, encompassing the universe, society, politics, and information about contemporary life, and are a global problem, felt at local levels, and will continue to be. Carbon dioxide, the heat-trapping greenhouse gas that has recently become the primary driver of global warming, remains in the atmosphere for thousands of years, and it takes time for the planet (especially the oceans) to respond to global warming. So, until we stop emitting greenhouse gases today, global warming and climate change will continue to affect future generations. As a United Nations, we are committed to making some changes to the climate.

What will be the extent of climate change? This will be determined by how we integrate and how our climate policies achieve their goals. Despite growing awareness of climate change, our global greenhouse gas emissions continue unabated. In 2013, the daily level of carbon dioxide in the atmosphere exceeded 400 parts per million for the first time in human history. This low level had not been seen for approximately three to five million years, since the Pliocene epoch.

Causes:

- **Power Generation (burning fossil fuels)**

A large portion of global emissions comes from burning fossil fuels to produce electricity and heat. Much of the electricity is still generated by burning coal, oil, or gas underground, producing carbon dioxide and nitrous oxide, potent greenhouse gases that trap and absorb the sun's heat. Globally.

- **deforestation**

Cutting down forests for carbon production, grazing, or other purposes generates emissions because, when cut, the forest releases carbon it didn't store. Nearly 10 million farmers are involved in forestry. Since carbon dioxide is not allowed, the impact is also determined by the ability to keep emissions out of the atmosphere. Deforestation, along with agriculture and other changes in work, accounts for nearly two-thirds of the world's greenhouse gas emissions.

(Figure 7)

Deforestation case

- **Use Transportation**

Most cars, trucks, ships, and airplanes run on fossil fuels. This makes transportation a major contributor to greenhouse gas emissions, particularly carbon dioxide. Road vehicles produce the largest share due to the combustion of petroleum products, such as gasoline, in internal combustion engines. However, emissions from ships and airplanes continue to grow. Transportation accounts for nearly a quarter of global energy-related carbon dioxide emissions. Trends indicate a significant increase in energy use in transportation in the coming years.

- **Factories**

emission of greenhouse gases such as carbon dioxide from the factories, while manufacturing products such as electronics, cement, steel, clothes, plastics, and other goods

Insecticides

Although insecticides prevent insects and Weeds from destroying crops, they also pollute the environment because they contain Harmful substances that cause climate change

Impact:

- More fires:

  Climate change makes the weather hotter and drier, which leads to the spread of fires, with the fires that spread faster and burn longer, putting many humans and animals at risk. The number of wildfires increased between 1984 and 2015 in the western United States and California, and the area burned by fires increased between 1972 and 2018.
- Melting ice sea:

  The effects of climate change appeared effectively at the two poles. The Arctic is heating up twice as fast as anywhere else on Earth, leading to the melting of glaciers and polar ice sheets. With the melting of the ice, the water of the ocean becomes exposed to sunlight. In just 15 years, the Arctic could be entirely ice-free in the summer
- Sea level rise:

  change in the climate leads to melting of the ice of the Arctic, raising the level of the oceans and the seas, causing floods that can destroy houses, land, and other things. Scientists predict an increase in sea level by as much as 6.6 feet by the end of the century, should we fail to curb emissions. According to the 2022 Sea Level Rise Technical

Report from the National Ocean Service, the United States will see a foot of sea level rise by 2050, which will regularly damage infrastructure, like roads, sewage treatment plants, and even power plants.

PROBLEM TO BE SOLVED

Egypt is facing a problem with its electricity demand, and it is relying too much on fossil fuels. This is why Egypt needs to find a way to use renewable energy. Solar and wind energy are options, but they are not perfect on their own. The main issue is that using one source of renewable energy is not efficient, so it is not reliable. A system that combines gear ratio and wind turbines can solve this problem. The way to make this system work better is to improve the gear ratio of the wind turbine.

Here is how it works:

• **Stage 1**: The gear ratio between the turbine axis and the generator is 1:2.

• **Stage 2**: The gear ratio in this stage is 1:3.

This setup makes the generator spin faster, which makes the energy conversion more efficient by 10%. This means that this power generation system is more reliable.

Negative Consequences (if not solved)

• Egypt will keep relying on fuels: If the gear ratio is not improved, hybrid systems will not work well, so Egypt will have to keep using oil and gas. This will increase emissions. Make the economy more vulnerable.

• Egypt will not have an energy supply: If the turbines are not designed well, they will not be able to produce energy consistently, so the solar-wind hybrid system will not be reliable when the wind is low or when it is cloudy.

• This will be bad for the economy: Inefficient turbines mean costs, more fuel imports, and fewer opportunities for local renewable energy industries.

• This will harm the environment: If emissions keep increasing, it will worsen pollution and climate change, which will threaten agriculture, public health, and sustainability.

Positive Consequences (if solved)

• Egypt will have a reliable energy supply: The two-stage gear ratio makes the turbine more efficient, so the hybrid system will produce energy more consistently, and there will be fewer blackouts.

• Egypt will reduce its greenhouse gas emissions: If the system is more efficient, Egypt will rely less on fuels, which will reduce emissions and help Egypt achieve its climate goals.

• This will help the economy grow and create jobs: efficient turbines will stimulate local manufacturing, installation, and maintenance jobs, and they will also reduce long-term energy costs.

OTHER SOLUTIONS HAVE ALREADY BEEN TRIED

Ras Gharib Wind Farm:

Overview:

Horizontal Axis Wind Turbines (HAWTs) are the most widely implemented technology for wind energy generation globally and in Egypt. In particular, the Ras Gharib region, located along the Gulf of Suez in the Red Sea Governorate, is considered one of the most suitable locations for wind energy exploitation due to its highly energetic and consistent wind regime. Large-scale wind farms have been established in this region to utilize its high wind potential and help reduce dependence on fossil fuels.

Figure 8
Ras Gharib Wind Farm

Climatological studies of Ras Gharib indicate that the region has mean annual wind speeds of 8.3-9.8 m/s and a high wind power density of approximately 360 W/m², making it well-suited for efficient wind energy conversion. Additionally, wind speeds fall within the optimal operational range of 5–17 m/s for approximately 73% of the year, ensuring continuous and reliable turbine performance.

Mechanism:

Horizontal Axis Wind Turbines (HAWTs) generate electricity by converting the kinetic energy of wind into mechanical energy, then into electrical energy. In Ras Ghareb, strong and consistent winds with average speeds of **8.3–**

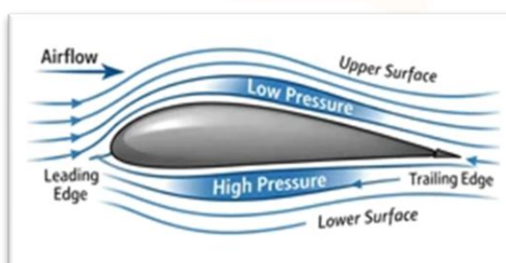


Figure 9
principles of aerodynamics

9.8m/s provide ideal conditions for efficient energy generation.

When air flows over the turbine's airfoil-shaped blades, a pressure difference is created due to Bernoulli's principle. The faster airflow above the blade results in lower pressure, while the slower airflow below creates higher pressure. This difference produces a **lift force** that rotates the blades, while drag remains minimal to maintain efficiency.

The rotating blades spin a shaft connected to a generator, converting mechanical energy into electrical energy through electromagnetic induction. The output depends on wind speed and rotational velocity.

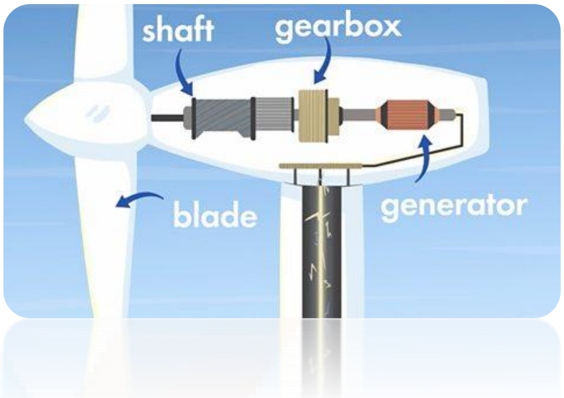


Figure 10
Describing rotating blades spinning a shaft connected to a generator

Ras Gharib's wind conditions provide a high energy density of about **360 W/m²**, with wind speeds in the effective range (5–17 m/s) for **~73% of the year**, ensuring continuous operation.

To ensure maximum energy capture, the turbine must always face the wind direction. This is achieved using a **yaw system**, which consists of a motor and sensors that detect wind direction and rotate the nacelle (the top part of the turbine) to align the rotor perpendicular to the wind. In smaller systems, a tail vane may perform this function passively, while in large-scale turbines, an active motor-driven yaw mechanism is used for precise control.

In addition, modern wind turbines include **braking systems** to protect the turbine from excessive wind speeds. One method is **aerodynamic braking**, in which the blades' pitch angle is adjusted to reduce lift and slow rotation. Another method uses mechanical braking systems, such as disc or hydraulic brakes, which are applied directly to the rotor shaft to stop or limit rotation during extreme wind conditions or during maintenance.

These systems are especially important in regions such as Ras Ghareb, where wind speeds can be very high. The braking system prevents structural damage, overheating of the generator, and mechanical failure due to overspeed.

Furthermore, the turbine's efficiency is limited by Betz's Law (59.3%), with practical ranges of **45–50% for large turbines** and **20–35% for small-scale systems.**

Points of Strengths:

1. High energy production capability.

Due to the high wind speeds in Ras Gharib (8.3–9.8 m/s) and a wind power density of approximately 360 W/m², HAWTs can generate sufficient torque to continuously drive the generator. These conditions allow the system to exceed the required energy output (≥150 J in 5 minutes) and ensure efficient conversion of wind energy into electrical energy.

2. Reliable operation under local wind conditions.

Wind speeds in Ras Gharib fall within the effective operating range (5–17 m/s) for about 73% of the year, indicating the turbine can operate for extended periods without interruption. This high operational availability makes the system suitable for continuous energy generation compared to other intermittent renewable sources.

3. Stable aerodynamic performance.

The dominant North-Northwest wind direction in Ras Gharib reduces the need for constant yaw adjustments. This minimizes mechanical stress on turbine components and improves stability, allowing smoother rotation and more consistent energy production.

4. High efficiency in large-scale systems.

Large HAWTs can achieve power coefficients of approximately 0.45–0.50, meaning nearly half of the wind's kinetic energy is converted into useful electrical energy. This makes them one of the most efficient renewable energy technologies when operating under optimal conditions.

Points of Weaknesses:

1. Dependence on wind availability.

The turbine cannot generate electricity when wind speed falls below the cut-in speed (typically 1.5–3 m/s), because the aerodynamic forces are insufficient to overcome the generator's resistance. This results in periods of zero energy production.

2. Fluctuating electrical output.

Variations in wind speed directly affect rotor rotational speed (RPM), causing unstable voltage and current output. This leads to inefficient energy delivery and potential losses in electrical components.

3. Requirement for wind alignment.

HAWTs must face the wind direction to operate efficiently, requiring a yaw system or tail vane. This adds mechanical complexity and introduces additional components that may increase maintenance requirements.

4. Reduced efficiency at the small scale.

Micro-scale turbines operate at low Reynolds numbers ($\sim 10^5$), where viscous forces dominate, increasing drag and reducing lift efficiency. As a result, small turbines produce less power compared to a large-scale system

Aswan High Dam:

overview:

Investing in massive infrastructure projects to increase availability in homes and businesses. The High Dam is considered the greatest and largest engineering project of the twentieth century. There were many steps in building the dam.

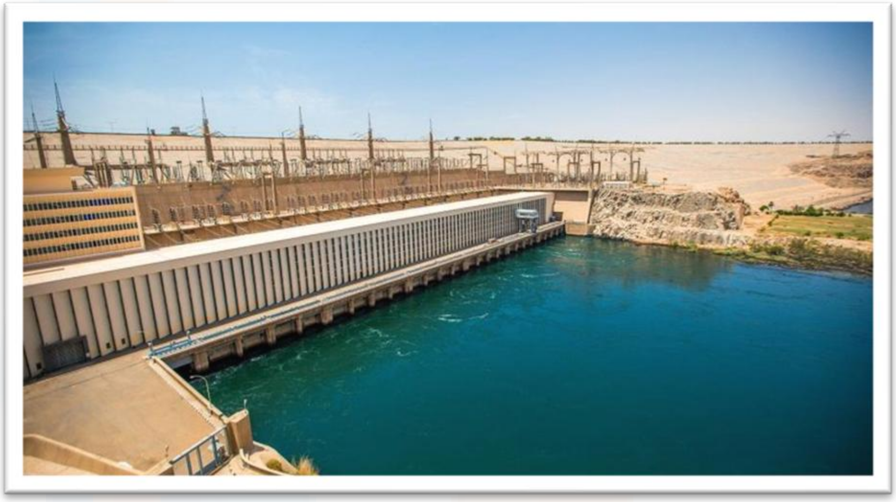


Figure 11
Aswan High Dam

The High Dam is 3.8 km long and took 11 years to complete. It was fully operational in 1971. Behind the dam, a large lake or reservoir was formed. This reservoir was formed. This reservoir, called Lake Nasser, as shown in Fig 4, is 650 km long and 35 km wide.

Figure 12
Lake Nasser and Aswan High Dam location

Mechanism:

The Aswan High Dam generates electricity by converting the potential energy of water into mechanical energy, which is then converted into electrical energy. Water from the Nile River is stored in Lake Nasser, which has a reservoir capacity of approximately **169,000 million cubic meters**.

When water is released from the reservoir, it flows through turbines, which spin and drive generators, producing electricity. The dam has a gross head of about **74 m**, allowing efficient energy conversion, and it generates approximately **10,000 GWh of electricity annually**.

The hydropower system uses **Francis turbines** supplied by Power Machines, comprising 12 units, each with a capacity of **175 MW**. As shown in fig(13)

Figure 13
Francis Turbines

The Francis turbine is a type of reaction turbine and is considered one of the first modern hydro turbines. It consists of a runner with fixed blades, typically nine or more, where water is introduced around the runner and flows through it, causing rotation. The system also includes key components such as a scroll case, wicket gates, and a draft tube, which regulate water flow and improve efficiency. These turbines are commonly used in medium- and high-head applications, such as the Aswan High Dam, and can operate in both vertical and horizontal configurations. Additionally, Electrosila supplied the electric generators used in the system to convert mechanical energy into electrical energy.

Points of Strengths:

1. It provides a majority of the energy needs of Egypt.

During the High Dam's usual operating year, about 15% of the country's total electrical supply is provided by this project. When it first came online, almost half of the available electricity flowed through the dam. The electricity generated by the dam is environmentally friendly, offers predictable cost structures, and is cost-effective to maintain. At total capacity, the 12 generators at the dam are each rated at 175 MW, allowing the facility to produce 2.1 GW of electrical energy.

2. Reliable and consistent energy generation.

Unlike solar and wind energy, hydropower is not affected by the environmental crisis, and it provides a constant energy supply.

3. Low operating cost.

Once built, the cost of maintenance and operation is relatively low compared to that of electric sources.

4. It does not pollute the environment.

Unlike fossil fuels, hydropower does not emit greenhouse gases when generating electricity.

Points of Weaknesses:

1- The project limited access to critical archaeological sites.

After the completion of the High Dam, over 20 sites and monuments were threatened by the spillages from Lake Nasser. Many sites had to be moved through UNESCO efforts to preserve them.

2- It changed how sediments flow to the sea.

The Nile River is famous for flooding its surrounding valley each year, providing cropland for local farmers.

3- The high dam forced over 1 million people to be relocated.

When Lake Nasser flooded lower Nubia due to the Aswan High Dam, up to 120,000 people had to be resettled in Egypt and Sudan. Another 70,000 Nubians in Sudan were resettled from Wadi Halfa, where their new home had a markedly different climate, making it difficult for them to adapt. They were eventually settled into 25 planned villages. Another 50,000 Nubians were moved up to 10 kilometers from the Nile into new village units as well.

Benban Solar Park:

Overview:

The Benban Solar Park is a large-scale photovoltaic (PV) power station located in Benban village in Aswan Governorate, in the Western Desert of Egypt, approximately **650 km south of Cairo and 40 km northwest of Aswan**. It is the largest solar park in Africa and one of the largest globally, with a total installed capacity of approximately **1650 MW (1.65 GW)**, corresponding to an annual electricity production of about **3.8 TWh**. The project consists of more than **30 independent solar power plants** developed by different companies, and it includes millions of photovoltaic solar panels (estimated over 6 million panels) distributed over a large area to maximize solar energy capture.

Figure 14
The Benban Solar Park

Mechanism:

The Benban Solar Park generates electricity using **photovoltaic (PV) technology**, which converts solar radiation directly into electrical energy. Each solar panel contains multiple photovoltaic cells made of semiconductor materials such as silicon.

When sunlight (photons) strikes the surface of these cells, it transfers energy to electrons within the semiconductor material. As shown in fig(15). This energy excites the electrons, causing them to move and create an electric current. This process is known as the **photovoltaic effect**.

The electricity generated at this stage is **direct current (DC)**. The DC electricity flows through conductive materials and is collected from multiple

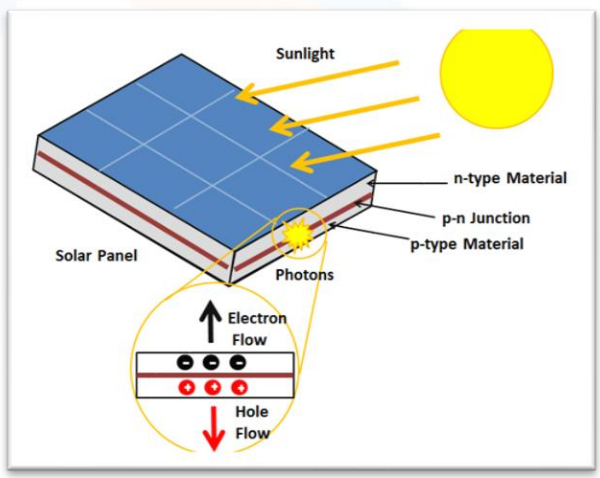


Figure 15
photovoltaic effect

panels connected in series and parallel configurations to increase voltage and current.

This electrical energy is then transferred to **inverters**, where it is converted from DC to **alternating current (AC)**, which is suitable for transmission through the national power grid. After conversion, transformers are used to increase the voltage level for efficient long-distance transmission.

Due to the high solar irradiance in Aswan (approximately **2200–2600 kWh/m²/year**), the system can operate efficiently for extended periods during the day, consistently producing large amounts of electricity.

Points of Strengths:

1. High large-scale energy production.

The use of millions of solar panels enables the system to generate approximately **3.8 TWh annually**, making it one of the most productive renewable energy projects in Egypt.

2. Clean and sustainable energy source.

The system produces electricity without emitting greenhouse gases during operation, reducing environmental pollution and supporting climate sustainability.

3. Modular and expandable design.

The system is divided into more than 30 independent units, enabling easy expansion, maintenance, and flexible operation.

4. Low mechanical losses.

Unlike wind or hydropower systems, solar panels have no moving parts, reducing mechanical wear and increasing system reliability.

Points of Weaknesses:

1. Dependence on solar radiation.

Electricity generation only occurs during daylight hours and stops completely at night, limiting continuous energy production.

2. Output variability due to environmental conditions.

Dust, clouds, and seasonal changes reduce solar irradiance, leading to fluctuations in energy output.

3. High initial capital cost.

The construction of the Benban Solar Park required a very large upfront investment (billions of dollars) to purchase millions of solar panels, inverters, transformers, and land for preparation, as well as grid infrastructure, making it difficult to implement without strong financial support.

4. Limited conversion efficiency.

Photovoltaic panels typically operate at efficiencies of about **15–20%**, meaning a significant portion of solar energy is not converted into usable electricity.

Éole Project:-

overview:

The **Éole wind turbine** in **Cap-Chat, Quebec, Canada**, is a unique and historically significant wind energy project. Located in the magnificent city of Cap-Chat in Gaspésie, it is a vestige of the beginnings of renewable energy research in Quebec. This 110-meter-high giant was an important step in the development of wind power.

Built in the mid-1980s, Éole is the tallest vertical-axis wind turbine in the world. Shut down in 1993, it no longer produces electricity and is accessible to the public for visits.

Figure 16
Éole Wind Turbine in Canada

Mechanism:

The type of wind turbine is a **vertical-axis wind turbine (VAWT)**, specifically a **Darrieus-type** turbine.

It has a **straight-bladed design**, unlike traditional horizontal-axis wind turbines (HAWTs).

With a height of **110 meters**, it remains the **tallest vertical-axis wind turbine** ever built.

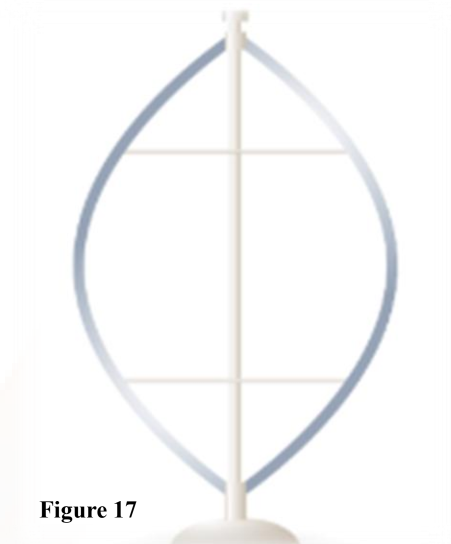


Figure 17

Darrieus shaped turbine (Eggbeater turbine)

- **Darrieus turbines** rely on **aerofoil-shaped blades** that rotate around a vertical axis.
- The **wind pushes the curved blades**, creating **lift** and causing them to rotate.
- Unlike horizontal-axis turbines, Éole does **not need to face the wind**, making it useful in areas with varying wind directions.
- The rotational mechanical energy is converted into — electricity by a generator at the base.
- Its blades are connected to **a central shaft** supported by **guy wires**, allowing it to capture wind from any direction

A problem with the Darrieus wind turbine is that it is not self-starting, so it uses its generator as a motor to start the rotor. As the wind increases the blade speed, power to the motor is cut off, and it begins operating as a generator.

Darrieus wind turbines were installed on early wind farms, but most have been taken out of commercial use because they are less efficient than HAWTs and require constant maintenance.

Points of Strengths:

1. **Omnidirectional Wind Capture** – Unlike horizontal turbines, Éole doesn't need to adjust to wind direction.

2. **Tall & Efficient in Low Wind Speeds** – Being 110 meters high, it could harness winds at greater altitudes.

3. **Easier Maintenance** – The generator is near the ground, making maintenance safer compared to horizontal turbines.
4. **Lower Noise Pollution** – Vertical turbines generally produce less noise than traditional wind turbines.
5. **Innovative Structure** – Éole was a pioneering design, proving the potential of large-scale VAWTs.

Points of Weaknesses:

1. **Structural Stress & Vibration** – The Darrieus design often experiences cyclic stress, leading to wear and tear.
2. **Inefficiency at High Speeds** – Vertical-axis turbines tend to be less efficient than horizontal ones, especially in strong winds.
3. **High Material Costs** – The large structure and guy-wire support system made it costly.
4. **Limited Commercial Viability** – It was an experimental project, and it didn't become a widespread solution.
5. **Decommissioned in 1993** – After a few years of operation, it was shut down due to technical and economic challenges.

RESEARCH

Topics related to the problem

Usage of alternative: Egypt is the third-largest emitter of greenhouse gases (GHGs) in the Middle East and Africa after Iran and Saudi Arabia, with GHGs reaching 351.96 $MtCO_2e$ in 2019. Egypt's energy sector is the top contributor to these emissions, responsible for 74% in 2019 (WRI, 2022; Enterprise, 2020).

Egypt is also the largest country in North Africa and the Arab region in terms of population, which has led to a growing demand for energy. This problem has further been exacerbated by the aggravation faster, he burden of the energy subsidy bill (IRENA, 2018). To meet the growing energy demand, the Egyptian government developed a strategy to diversify energy sources known as the "Integrated and Sustainable Energy Strategy to 2035", to ensure the continued security and stability of the country's energy supply. This strategy entails enhancing the role of renewable energy and energy efficiency, while targeting diversification of the energy mix by increasing reliance on renewable energy sources, especially wind and solar energy (CAPMAS, 2016)

Egypt adopted the first renewable energy strategy in 1982, aiming to generate 5% of its electricity from renewable sources in 2000 (EU, 2015). In 2013, the government adopted an integrated strategy for sustainable energy from 2015 to 2035 through a project funded by the European Union. In October 2016, the strategy was approved by the Supreme Council of Energy (IRENA, 2018).

To achieve sustainability in the energy sector, the government has developed fiscal, monetary, and trade policies to develop renewable energy sources and encourage investment. One notable fiscal policy tool is taxes, with the capital components of renewable energy being subject to a value-added tax (VAT) of only 5% instead of 14% according to the

VAT law (IRENA, 2018). In February 2017, the Central Bank added the renewable energy sector to the initiative of medium-sized companies and enterprises issued in 2016 (a 200 billion EGP initiative) to encourage banks to finance small and medium-sized companies and enterprises. (Abbas, 2019).

Additionally, the government has taken several measures to encourage the participation of the private sector in renewable energy projects. The most important step in this regard was the legislative amendments made to remove obstacles to investment in this field; most notably, the approval of the feed-in tariff system in September 2014 to encourage the production of electricity from renewable sources (NREA, 2020).

Recycling:
Overflowing Landfills Without recycling, plastic waste would pile up in landfills at an alarming rate. Within a year, landfills would be overwhelmed, leaving no space for additional waste.

Increased Pollution:
Discarded plastic would likely end up in oceans, rivers, and other ecosystems, leading to severe environmental damage. Marine life would be particularly affected, as animals could ingest plastic debris, causing starvation, injury, or death.

Harm to Wildlife:
Wildlife, both on land and in water, would suffer greatly. Animals may ingest or become entangled in plastic waste, leading to injuries or fatalities.

Loss of Valuable Resources

Plastic recycling reduces the need for virgin plastic production, which requires fossil fuels like petroleum. If recycling stops, the demand for raw materials will surge, depleting natural resources faster and increasing greenhouse gas emissions from production processes.

Increased Carbon Footprint:

Producing new plastic releases more carbon dioxide than recycling existing materials. A halt in recycling would significantly increase global carbon emissions, exacerbating climate change and global warming.

Global Waste Crisis:

The global plastic waste crisis would escalate. By not recycling, the world would generate an estimated 300 million tons of additional plastic waste in a year. Countries with inadequate waste management infrastructure would be particularly overwhelmed.

Topics related to the solution

Horizontal wind turbine:
HAWT has high generating capacity, improved efficiency, variable pitch blade capacity, and a tall tower to capture large amounts of wind energy. HAWT usually has two or three rotor blades. A turbine with two rotor blades is often used in downwind installations where the rotor is downwind of the tower. It is faster and cheaper, but it flickers more than the rotor with three blades and is less efficient. Three-blade rotors operate more smoothly and are therefore less disturbing. Advantages of HAWT include ability to pitch the rotor blades in a storm so that damage is minimized; collection of the maximum amount of wind energy by allowing the angle of attack to be remotely adjusted; stability, since blades are to the side of the turbine center of gravity; access to stronger wind in sites with wind shear and placement on uneven land or in offshore locations using tall tower; and most HAWTs are self-starting. Now we will speak about the 4 main advantages of the horizontal wind turbine:

High Power Output:

Horizontal-axis wind turbines are generally built to have a capacity ranging between 2 and 8 MW, depending on the usage. While the output of a wind turbine depends on the turbine's size and the wind speed, an average onshore wind turbine with a capacity of 2.5 – 3.0 MW can produce more than 6 million kWh in a year, which is enough to supply 1,500 average EU households with electricity.

High Efficiency:

For any type of energy conversion, there is always energy loss. How to improve energy conversion efficiency is one of the biggest focuses of product development in the wind energy industry. Currently, horizontal-axis wind turbines have the highest efficiency. They can transform 40 to 50% of the received wind power into electricity.

High Reliability:

Being the dominant wind turbine model for decades, research and development of horizontal-axis wind turbines are already mature. Not only are existing products on the market reliable, but the application and usage of horizontal-axis wind turbines are also thoroughly explored.

High Operational Wind Speed:

Due to the height of the rotors, horizontal-axis wind turbines can receive wind at a greater speed. This means they are more likely to operate at higher wind speeds, which helps them provide optimal performance. Since the air flow at such a height is relatively stable, horizontal-axis wind turbines enjoy more consistency in wind and thus in power output.

CHAPTER II

Generating and defending a solution: -

SOLUTION AND DESIGN REQUIREMENTS

Requirements

System Efficiency

In the beginning, students must clearly define the input of energy, which represents the total amount of energy supplied to the system. This could come from various sources depending on the design, such as electrical power, fuel combustion, or renewable resources like solar or wind energy.

Secondly, the output energy refers to the useful energy delivered by the system. This might take the form of motion, electricity, or heat that can be practically utilized.

The relationship between these two is expressed through efficiency, which indicates how much of the input energy is successfully converted into output rather than being wasted. In real systems, some energy is always lost due to factors such as friction, resistance, or heat loss. Therefore, identifying and understanding these losses is an important part of the analysis.

Students should not only calculate efficiency but also explain the factors that influence it and how they impact the system's overall performance.

Required Energy Output Performance

The prototype must be capable of generating a measurable amount of energy to demonstrate its effectiveness. Specifically, it should produce at least 150 Joules of energy within a period not exceeding 5 minutes

This requirement ensures that the system is not only conceptually valid but also practically functional. It pushes students to design a system that delivers consistent and reliable output within a limited time frame.

To verify this, students should apply appropriate measurement methods depending on the nature of their system. Whether through direct electrical measurements or calculated physical quantities, the process used to determine energy output must be clear and well-explained.

Performance Enhancement and Efficiency Improvement

Another important requirement is to improve the system after its initial testing phase. Students must demonstrate that their modified prototype achieves at least a 10% increase in efficiency.

This improvement should be supported by clear evidence, including recorded data before and after modifications. Additionally, the increase in efficiency should be accompanied by a noticeable improvement in the total energy produced.

Students are expected to conduct tests under varying conditions and observe how these changes affect system performance. Important variables to consider include:

Environmental conditions, such as temperature or wind speed, may affect energy generation or system efficiency

Type of energy source or fuel, which can influence both output and efficiency

Design-related factors, including material selection, component quality, and system configuration

The data collected should be organized and analyzed carefully. Graphs and tables can be used to highlight trends and relationships. More importantly, students should interpret their results and explain why certain changes lead to better or worse performance.

Solution

- Technical Mechanism and Scientific Justification

The wind component utilizes the principle of Kinetic Energy Conversion. As wind flows over the aerodynamically optimized blades of the horizontal turbine, it creates a pressure difference that rotates the internal generator, converting mechanical torque into electrical energy. Simultaneously, solar panels operate through the Photovoltaic Effect, where semi-conductor materials absorb photons and release electrons, creating a direct current (DC). The scientific novelty of our solution is the "Redundant Power Management System." By utilizing two independent battery storage units, we effectively isolate the charging circuits. This prevents the "bottleneck" effect, where one source might underperform (due to clouds or low wind) and drag down the overall efficiency of the system. Furthermore, it provides a critical fail-safe mechanism: if the solar circuit encounters a technical fault, the wind circuit remains fully operational, maintaining a continuous power flow to the load.

- Climate Compatibility in Egypt

The Global Horizontal Irradiation (GHI) levels in Egypt are among the highest in the world, making it an excellent location for solar energy. The intermittent nature of sunlight adds to the challenge of using solar as a source of energy. Wind energy has been incorporated into our system to help eliminate that challenge, and the location of Egypt on the Red Sea and Mediterranean coast creates an opportunity for year-round, consistent wind and solar to produce energy together. This hybrid system creates a "Micro-Grid" that creates "Stability" for use in off-grid areas such as rural and desert areas that don't have access to the national grid.

- The following are requirements and limits to the design.

To ensure that the project can be built, there are two requirements. The system must be easily transportable, to meet the requirement of being "Modular" and "Scalable", so that it allows for additional capacities in the future. In addition, safety regulations are outlined in the form of implementing both a Charge Controller to avoid overcharging the batteries and an Automatic Braking System for the wind turbine, which, in high winds or sandstorms, will prevent damage to the structure from strong winds.

In terms of operational requirements, we are specifying that the solar panels need to be able to achieve a minimum of 18% efficiency and, in addition, to have a lower-than-expected cut-in wind speed, so they will still operate in low-resource regions.

Design Requirements and Constraints

To ensure the feasibility of the project, several criteria must be met. The system must be "Modular" for easy transportation and "Scalable" to allow for future capacity increases. Safety constraints require the use of a Charge Controller to limit battery overcharging and an Automated Braking System on the turbine to prevent structural failure in high winds or sandstorms. Our functional requirements also include a minimum efficiency of 18% for solar panels and a low cut-in wind speed, so the system can still function in areas with limited resources.

SELECTION OF SOLUTION

The main idea of our solution is to be able to generate at least 150 joules of energy within a maximum amount of time of 5 minutes. The prototype volume must not exceed 0.25 m^3. The enhancement we made on our prototype must provide at least a 10% improvement on the original output.

The main source that will provide us with the needed joules and the original output will be the horizontal wind turbines. The horizontal wind turbine can absorb 40% of wind energy and convert it into electric energy. This will prove to be very useful and will allow us to very easily be able to generate the needed 150 joules very quickly. The horizontal wind turbines operate at high speed, which means they can receive wind energy very quickly, which then means they can generate electricity at very high speed, which means great energy/power output consistency, which will help us even more in trying to achieve the needed 150 joules.

Now, as for our main energy converter, we will be using the DC generator. A DC generator is a good choice for small-scale wind turbine systems as they are reliable, can operate at low rotational speeds, and provide good efficiency, especially in light wind conditions.

Now, as for the material that can be recycled, we will use wood as we will use for 2 things. First, we will use it as a wooden box that we will place all of our gears inside. Second, we will use it as the base of our prototype by connecting it using nails.

Now, as to how we are going to achieve the 10% enhancement needed, we will be using the gear ratio. First, we will start by using simple yet good gears for both the wind turbine and the generator. These gears will be enough to allow us to produce the needed 150 joules. At this point, the gear ratio would be 1:3. Afterwards, we will then switch these gears for better ones, which would bring our ratio up to 1:4. By doing this, we can achieve the needed 10% enhancement.

SELECTION OF PROTOTYPE

The design presented above represents a prototype of the three-dimensional horizontal-axis wind turbine (HAWT) that has three aerodynamic blades in (NACA 0018) design, fixed to a hub, which connects blades with a shaft, a gearbox, and a DC generator. Mechanical energy received due to rotation is converted to electrical energy via the generator, whereas the use of the gearbox that consists of 2 gears, 1 large, which is fixed on the shaft axis, and 1 small, which is fixed on the generator, in the ratio of (3:1) is meant to increase the rotational speed to improve efficiency.

Figure 18
Our 3D design of prototype

While constructing the prototype, several key factors were considered to achieve two main objectives: maximizing efficiency and meeting the design requirements.

The first design choice was selecting a three-blade HAWT configuration, as it provides an optimal balance between aerodynamic efficiency, rotational stability, and cost. The blade length was carefully chosen to increase the swept area, as its length is 42 ± 0.5cm, allowing the turbine to capture more wind energy and improve overall performance.

The second design feature is the gearbox system (4:1 ratio), which was integrated to enhance the rotational speed of the generator. Since small-scale turbines produce low rotational speeds under moderate wind conditions, the gearbox increases generator speed, resulting in higher voltage and energy output, ensuring that the system meets the requirements, which is to increase the output with 10%.

Third, the use of a 12V DC generator has been chosen as the method for the transformation of mechanical energy to electrical energy. It is obvious that the efficiency of DC generators is determined by the rotational speed; therefore, it is important to optimize turbine speed.

Fourth, the support structure for a wind turbine should not exceed a volume of 0.25 m3. For this purpose, the dimensions of the wind turbine are equal to 110 ± 0.5 cm x 70 ± 0.5 cm x 20 ± 0.5 cm, which results in the total volume of about 0.154 m3 ± 0.002 m3. A wooden nacelle, made from recycled material, will be used to accommodate and protect the generator and other mechanical elements.

It means that the proposed solution provides sufficient efficiency, stability, and the possibility to generate the required electrical energy (at least 150 J within 5 minutes) under realistic wind conditions.

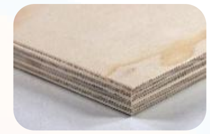
CHAPTER III

Constructing and Testing Prototype: -

MATERIALS

&

METHODS

Item	quantity	Usage	Cost	Sourcee	picture
Roll bearing	7	Reduces friction massively, allowing the turbine to spin.	420 L.E	Local shop	
Fast adhesive	1	Gluing different parts of the prototype	105 L.E	Local shop	
Acrylic sheets	1	Creating gears since it causes less friction	0 L.E	FabLab	
PLA+ Roll	1	Printing the turbine blades and tower using a 3D printer	0 L.E	FabLab	
12v DC motor	1	Generate electricity	300 L.E	Local shop	
Plywood	1	Creating the base box and the blades' holders	0 L.E	**Recycled**	
Copper wire	2m	Used wires to connect the motor to the multimeter	15 L.E	Local shop	

Preconstruction:

1. We picked out our detailed 3D design of the prototype, as shown in Figure 2 on Fusion, and then visualized it and printed it out.

2. We prepared the desired tools and materials.

After the 3D model materials had arrived, we started constructing:

1. Beginning with the base, MDF wood was used to cut a wooden base of dimensions 30cm (±0.5), 15cm (±0.5), and 5cm (±0.5) with a circular hole of a diameter of 2.5cm (±0.1) at the center of the lower surface of the base.

2. We connected the blades with glue, as each blade was separated into 3 pieces.

3. At the end of each blade, we then placed the bearing.

4. We added the gears under the bearing so that we can freely move the blades' direction

5. We placed each end of the blade in its designated placement on the hub as shown in Figure 3.

6. We placed the tip and head of the hub on top of the blades so that they were firmly secured and wouldn't fall off.

7. We added a hole to the wooden box and then inserted the shaft of the wind turbine.

8. We added a gear that connected the wind turbine's gears with the generator

9. Afterwards, we placed the wooden box and blades on a pole that we had also 3D printed

10. We inserted the pool onto a wooden base and made sure it was connected by connecting the pole base with the wooden base using nails

Our prototype was now fully built and complete.

We had 2 main design requirements that we needed to achieve.

First: achieving at least 150 joules of energy in 5 minutes. We achieved the first design requirement easily by conducting 156.27±3 joules

Second: making some improvements to the prototype, which would result in at the very least a 10% improvement to the original output in 5 minutes. We achieved the second design requirement by making a massive 109.8% enhancement.

Third: using at least one material that can be recycled. We achieved the final design requirement by using wood as our base and a wooden box for the nacelle.

TEST PLANS

Testing the prototype is one of the last steps in EDP, and it is so vital that it shows whether our prototype achieved the design requirement or not. There are some design requirements that our test plan is based on, **which are the following:**

Figure 19
A figure that describes the test plan

- Being able to achieve at least 150 joules of energy in 5 minutes.
- The volume of the prototype does not exceed 0.25m3.
- Successfully using a recycled material, so it's eco-friendly
- Making some kind of improvement to the prototype, which would result in at the very least a 10% improvement to the original output in 5 minutes.

There are some steps we followed in testing the prototype, **which included the following:**

Figure 20
Another figure describes the test plan

- Choosing the best wind conditions we can get.
- Putting the prototype on the 5th floor of a building.
- We made sure to always be near the prototype so that if the blades spin too quickly, we can stop it by changing its direction so it isn't directly facing the wind, so it wouldn't break.
- Connecting the positive and negative wires from the generator to a multimeter.
- Taking the average of the current intensity and the potential difference after each minute.• Calculating power and energy.

DATA COLLECTION

During the testing phase, we aimed to verify whether the turbine produced the needed amount of current before and after the enhancements, and to determine if further adjustments were needed. Throughout this process, we faced several negative results, which guided us to refine and improve the design and the mechanisms of some parts. After applying these modifications, the final test confirmed that the turbine generated an optimal amount of energy, measured accurately using both voltage and current readings.

Negative results:

During the first test, we didn't achieve the needed amount of voltage because the blade angles weren't aligned to perfectly face the wind.

We tried to use a PVC pipe for the tower of the turbine, but we couldn't adjust the measurements, especially the slope of the pipe that we needed, so we used a 3D printed tower instead.

-Moreover, the gears were damaged by two factors: the first factor is that the nacelle is small relative to the gears, and there was a friction force between the nacelle and the gears.

Second factor, making the axis between the rotor hub short so that the gear wouldn't fit in the axis well and wouldn't reach the generator.

Figure 22
A figure describes the highest voltage before enhancement stage

Figure 21
A figure describes the highest ampere before enhancement stage

Positive results:

In the positive results, we started by improving and enhancing the negative
First, we started by interlocking the blades in the rotor hub to make it resist higher wind speeds. Then, we aligned the blades to face more

wind and put the pitch control to make the blade angles controllable. For the factors that damage the gears, we made the nacelle bigger and used a wooden box instead of the 3d printed nacelle, then we 3D printed a taller axis to hold the gears well. And as shown in "Table 1".

Figure 24
A figure describes the highest Voltage after enhancement stage

Figure 23
A figure describes the highest ampere after enhancement stage

Table 1

Showing the positive results we achieved before the enhancement stage at 4m/s speed of air

Time	60	120	180	240	300
Current intensity(amp) ($\pm 0.002\ A$)	0.087	0.098	0.107	0.112	0.119
Potential difference(v) ($\pm 0.2\ V$)	3	4.6	5.2	5.9	6.2
Power(Watt)	0.261	0.4508	0.5564	0.6608	0.7378
Energy(J)	78.3	135.24	166.92	198.24	221.34

In the enhancement phase which we changed the gear ratio from 1:3 to 1:4, giving us a better mechanical advantage, and that enhancement generated more electric current, which leads to producing more amount of joules as shown in “Table 2."

Table 2

Showing the positive results we achieved after the enhancement stage at 8m/s speed of air

Time	60	120	180	240	300
Current intensity(amp) $(\pm 0.002\ A)$	0.112	0.138	0.174	0.194	0.205
Potential difference(v) $(\pm 0.2\ V)$	3.9	5.8	7	7.8	8.7
Power(Watt)	0.4368	0.8004	1.218	1.5132	1.7835
Energy(J)	131.04	240.12	365.4	453.96	535.05

CHAPTER IV

Evaluation, Reflection, and Conclusions: -

DISCUSSION AND ANALYSIS

Intro:

Egypt is currently facing serious environmental problems. Egypt's rising electricity consumption and increased use of fossil fuels underscore the need to expand alternative energy resources. Among the Egyptian Grand Challenges, improving renewable energy efficiency is crucial for achieving sustainability, pollution control, and climate change adaptation. A horizontal-axis wind turbine is ideal. A 156.27$\pm$3 J in 5 min was produced in the first stage. After improving our prototype, our results increased with 109.8% as it produced 327.8832$\pm$3 J on average in 5 minutes. Thus, we met the design requirements of producing sufficient energy and enhancing it; the prototype's volume doesn't exceed the limit of 0.25 m^3, as it's 0.154 m^3. It is both eco-friendly and user-friendly. It uses recycled material, and it doesn't make noise.

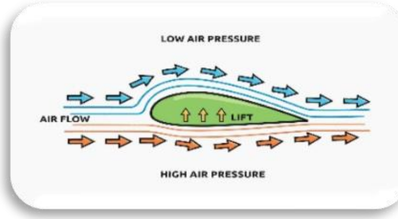
Design and Mechanism:

Horizontally axis wind turbines (HAWTs) have been known in Europe since the 12th century, in the form of mechanical windmills. The very first HAWT capable of generating electricity was created in 1887 by James Blyth. Further improvements made by Poul la Cour were responsible for creating today's efficient HAWTs.

(Figure 25)

A figure that describes 3D-design of our project

In our project, a three-bladed horizontal axis wind turbine was chosen, as shown in Figure 25, as the most suitable option in terms of optimizing the ratio between aerodynamic efficiency, stability, and economic aspects compared with different types of blade arrangements. The length of each blade has been fixed at 42 cm to optimize the swept area in order to maximize the captured wind power. Based on Betz's Law, the highest efficiency that a wind turbine can achieve is 59.3%, which means that increasing the rotor swept area will increase the amount of extractable energy from the wind (Betz, 1920). The overall dimensions of the prototype are 110 ± 0.5 cm x 70 ± 0.5 cm x 20 ± 0.5 cm, resulting in a total volume of 0.154 m^3, which does not exceed the design constraint of 0.25 m^3. This confirms that the project satisfies the required limitations while keeping its performance effective.



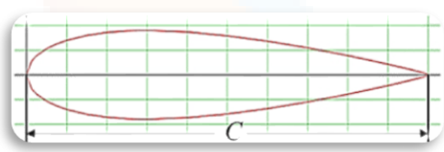
(Figure 26)

A figure that explains Bernoulli's principle causes the lift force

The horizontal-axis wind turbine includes 3 airfoil blades (NACA 0018), installed on the rotor and attached. The rotor rotates together with a shaft that delivers the mechanical energy from the rotor to the generator. When the flow of wind hits the blades, the rotor starts rotating due to differences in pressure where the Air moves faster over the curved upper surface, and this creates low pressure.

Air moves more slowly below, resulting in high pressure, as shown in Figure 26, thus producing mechanical energy transmitted to the shaft and further converted into electricity. For improving efficiency, pitch control is applied. By adjusting the position of the blades to wind changes, one makes sure that the blade maintains the optimal angle of attack within the range from 5° to 8°. As already mentioned, the selected airfoil (NACA 0018) has a symmetric cross-section, as shown in Figure 27, and allows for lift generation at positive angles of attack, providing stable operation.

In conclusion, the rotor delivers rotational energy to the main shaft of the turbine.



(Figure 27)

A figure that explains NACA 0018 airfoil profile

Principles and laws:

The operation of the horizontal-axis wind turbine is based on several fundamental physical principles.

First, Bernoulli's principle explains the generation of lift. It states that an increase in the speed of a fluid occurs simultaneously with a decrease in pressure or potential energy. This relationship is expressed by Bernoulli's equation:

$$P_1 + \frac{1}{2}\rho v_1^2 + \rho g h_1 = P_2 + \frac{1}{2}\rho v_2^2 + \rho g h_2$$

As air flows over the turbine blade, it moves faster over the curved upper surface and slower beneath it. According to Bernoulli's principle, this creates a pressure difference, resulting in a lift force that drives the rotation of the blades (NASA, n.d.; Serway & Vuille, 2011).

The magnitude of the lift force depends on wind speed and blade design and is expressed by:

$$L = \frac{1}{2}\rho A v^2 C_L$$

Where $(L)$is the lift force, $(\rho)$is air density, $(A)$is the blade area, $(v)$is wind speed, and $(C_L)$is the lift coefficient. This equation shows that lift increases with the square of wind speed and depends on the airfoil shape and angle of attack (NASA, n.d.; Serway & Vuille, 2011).

The angle of attack is a critical factor affecting performance. Maintaining it within the optimal range of 5°–8° maximizes the lift-to-drag ratio, ensuring efficient energy conversion while preventing aerodynamic stall (U.S. Department of Energy, n.d.).

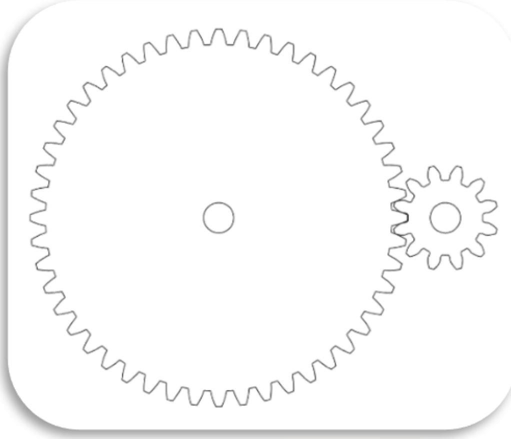
Additionally, the rotation of the turbine depends on torque, which is given by:

$$\tau = r \times F$$

This indicates that increasing the distance between the blade and the shaft increases the torque, allowing the turbine to rotate more effectively and generate more power (Halliday et al., 2014).

Gears' mechanical advantages:

Gears are essential components for transmitting power and motion between different parts of a machine. When two gears are connected, and the first gear (the driving gear) has more teeth than the second gear (the driven gear), the smaller gear will rotate faster. We used spur gears because they are the simplest, most common, and highly efficient (up to 99%) cylindrical gears with straight teeth cut parallel to the axis of rotation. The gear-tooth ratio can indicate an increase in speed. If the driving gear has 20 teeth and the driven gear has 5 teeth, the driven gear will rotate four times for every full rotation of the driving gear. Our gear-tooth ratio after enhancing is (4:1). The larger gear, which is 48-tooth, acts as the driving gear. As shown in Figure 28, which is fixed on the shaft, transferring motion to the smaller driven gear, which is 12-tooth. That is fixed on the generator. The equation: $\frac{N_1}{N_2} \times \frac{N3}{N4} \times \frac{N5}{N6} \times \ldots \frac{Nx}{Nx+1}$



(Figure 28)

A figure that explains our gears

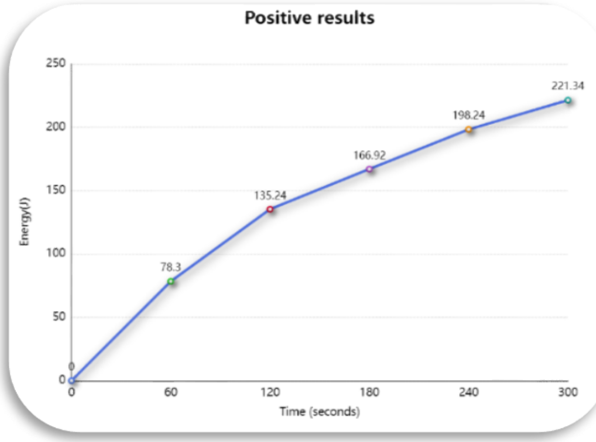
N: number of teeth

Results analysis:

In this project, a multimeter was used to measure the voltage and the current intensity of the energy produced by the Horizontal wind turbine. These Measurements were used to calculate the number of watt-seconds produced. making sure that the prototype met the specified design requirement. Multiple scientific laws were applied:

Power($\boldsymbol{W}$) = electric current intensity(A) × potential difference($\boldsymbol{V}$)
Energy(J)= Power($\boldsymbol{W}$) × $\boldsymbol{Time}$(S) (Halliday et al., 2014; Serway & Vuille, 2011).

Within a span of 5 minutes, the highest voltage recorded was 6.2 V, while the average voltage output was 4.98±0.2 V. The maximum current intensity output was 0.119 A, while the average current was 0.1046±0.002 A. With these results, the power output was calculated to be 0.7378 Watts. The prototype generated avg watt sec is 156.27±3 Ws (joules) in 5 min, as shown in Figure 29. As such, it can be concluded that the prototype meets the design requirement that states the production of more than 150 J of energy output in 5 minutes during the first stage before the enhancement.

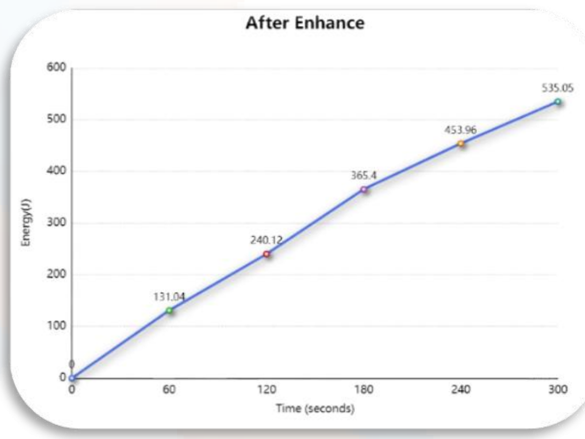


(Figure 29)

A graph describes the produced energy before enhancement

After enhancement:

Within 5 minutes span, the highest voltage recorded was 8.7 V, while the average voltage output was 6.64±0.2 V. The maximum current intensity output was 0.205 A, while the average current was 0.1646±0.002 A. With these results, the power output was calculated to be 0.7378 Watts. The prototype generated avg watt sec is 327.8832±3 joules in 5 minutes, as shown in Figure 30. As such, our results increased by 109.8%, and it can be concluded that the final prototype meets but exceeds the design requirements that state that the production energy must be increased by 10%.



(Figure 30)

A graph describes the potential difference after the enhancement stage

Our final production was more than 165 J of energy output in 5 minutes.

Our results demonstrate that Egypt’s environment effectively supports wind turbine electricity production. As illustrated in the results.

RECOMMENDATION

As time passes, technology improves, and a fact etched in mind that no project is perfect, and there must be room for improvement, so here are some recommendations for the teams that are going to work on our project or improve it:

- Using a metallic gearbox should be used in a wind turbine because it can handle very high torque, continuous loads, temperature changes, and long-term fatigue.
- Interlocking the turbine tower with the base, which is going to make the tower more stable and resist higher air speed without any deflection.
- Using carbon fiber in the axis that connects the blades and the gears. Carbon fiber is an ultra-strong, lightweight synthetic material composed of thin crystalline filaments of carbon atoms. And these properties would help the turbine to be more efficient.
- Using sensors and electronics. Sensors will make the turbine simple and support it in generating more. Example using a pitch angle sensor that helps in identifying the best angle for the blades to produce more electric current.
- Using braking pads. Braking pads are essential to stop the turbine and prevent it from damage, and to perform mechanical repairs.

LEARNING OUTCOMES

Learning Outcome (LO)	Concept	Content	Relation
Earth Science (ES.1.09)	Energy resources	We learnt the different types of energy production and their sources	Understanding the different forms of energy helped us realize the importance of renewable energy and how to exploit it.
Earth Science (ES.1.10)	Potential renewable energy	We learnt a brief explanation of the most common types of renewable energy.	This gave us a brief view of how electricity is generated from wind and how wind turbines work.
Mechanics (ME.1.06)	Resultant and Resolution of forces, Lami's rule, and the triangle of forces	We learnt how to deal with forces on the same plane, resolve them, and determine the final direction of the forces.	Learning the ability to resolve and understand forces acting on an object helped us determine the forces acting on the wind turbine's rotors and make sure the resultant force pushes the turbine, so it rotates.
Mechanics (ME.1.07)	Friction	We learnt about static and dynamic friction and how they act against the object.	By learning how to use friction and include it in our calculations, we were able to calculate the amount of air friction against the wind turbine and how it affects its rotation speed.
Mechanics (ME.1.09)	Centripetal force and centripetal acceleration	We learnt how to calculate the force acting on an object moving in a circle and calculate its	Learning this helped us calculate the amount of acceleration our wind turbine will gain by the centripetal force acting

		acceleration along the center of the circle.	on it, and what speeds it can reach.
Chemistry (CH.1.09)	Physical properties	We learnt the different physical properties of different materials	This helped us choose the materials for our prototype since we needed specific properties like lightweight or high malleability.
Physics (PH.1.09)	Bernoulli's equation and fluid properties	We learnt how fluids act when in motion and a few equations to help us analyze them, like Bernoulli's equation.	Bernoulli's equation is the primary principle that explains how wind turbines work; learning it helped us understand our solution better.
Physics(PH.1.08)	Fluids, types of pressure, and pressure difference.	We learnt what fluids are, types of pressure, how pressure functions, and pressure difference and force	This understanding helped us see how air moves from high pressure to low pressure, which causes our turbine to spin and generate electricity

Chemistry (CH.1.14)	Electrochemical cells and Galvan cells	We learnt how batteries work, and discovered that they are a type of electrochemical cell, especially the Galvan cell. These cells generate electricity through spontaneous chemical reactions, converting chemical energy into electrical energy.	Understanding how to store electricity through electrochemical cells, such as batteries. These batteries operate through a process called redox, where chemical reactions transfer electrons, generating and storing electrical energy.
Physics(PH.1.10)	Law of conservation of energy	We learnt about the Law of Conservation of Energy, which states that energy cannot be created or destroyed—only transformed from one form to another.	We learned that electrical energy isn't created from nothing and that it was converted from mechanical to electric energy through a generator.

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